

Φάση Ε: Model Evaluation & Σύγκριση

Υπεύθυνος: ML Engineer

Μετρικές:

- Accuracy
- Precision / Recall
- F1-Score
- ROC-AUC
- Confusion Matrix

Οπτικοποιήσεις:

1. Confusion Matrix (heatmap)
2. ROC Curve
3. Συγκριτικά bar charts
4. Πίνακας σύγκρισης

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# Φάση Ε & ΣΤ: Model Evaluation, Σύγκριση & Συμπεράσματα (ΠΛΗΡΗΣ ΚΑΛΥΨΗ + F1-SCORE)
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from pyspark.sql import SparkSession
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.metrics import (accuracy_score, precision_score, recall_score,
                             f1_score, confusion_matrix, roc_curve, auc,
                             precision_recall_fscore_support,
                             classification_report)

print("Εκκίνηση SparkSession...")
spark =
SparkSession.builder.appName("Evaluation_Metrics_Final").master("local[*]").getOrCreate()

print("Φόρτωση προβλέψεων (Pandas)...")
df_rf = spark.read.parquet("../data/rf_predictions.parquet").toPandas()
df_svm = spark.read.parquet("../data/svm_predictions.parquet").toPandas()
df_nb = spark.read.parquet("../data/nb_predictions.parquet").toPandas()
df_mlp = spark.read.parquet("../data/mlp_predictions.parquet").toPandas()
df_gbt = spark.read.parquet("../data/gbt_predictions.parquet").toPandas()
df_lr = spark.read.parquet("../data/lr_predictions.parquet").toPandas()

# Λεξικό για πλήρη αποθήκευση μετρικών ανά κλάση και συνολικά
metrics_dict = {
    "Model": [],
    "Acc (Total)": [],
    "P (Class 0)": [], "R (Class 0)": [], "F1 (Class 0)": [],
    "P (Class 1)": [], "R (Class 1)": [], "F1 (Class 1)": [],
    "ROC-AUC": []
}
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}

def evaluate_model_full(df, model_name, prob_col="probability"):
    y_true = df['stroke'].values
    y_pred = df['prediction'].values

    # ROC-AUC
    y_prob = np.array([v[1] for v in df[prob_col]]) if prob_col == "probability"
    else np.array([v[1] for v in df[prob_col]])
    fpr, tpr, _ = roc_curve(y_true, y_prob)
    roc_auc = auc(fpr, tpr)

    # Μετρικές ανά κλάση (Precision, Recall, F1)
    precisions, recalls, f1s, _ = precision_recall_fscore_support(y_true, y_pred,
labels=[0, 1], zero_division=0)
    acc = accuracy_score(y_true, y_pred)

    # Προσθήκη στο λεξικό
    metrics_dict["Model"].append(model_name)
    metrics_dict["Acc (Total)"].append(acc)
    metrics_dict["P (Class 0)"].append(precisions[0])
    metrics_dict["R (Class 0)"].append(recalls[0])
    metrics_dict["F1 (Class 0)"].append(f1s[0])
    metrics_dict["P (Class 1)"].append(precisions[1])
    metrics_dict["R (Class 1)"].append(recalls[1])
    metrics_dict["F1 (Class 1)"].append(f1s[1])
    metrics_dict["ROC-AUC"].append(roc_auc)

    cm = confusion_matrix(y_true, y_pred)

    # Εκτύπωση αναλυτικού classification report
    print(f"\n{'='*20} {model_name} {'='*20}")
    print(classification_report(y_true, y_pred, target_names=["Class 0", "Class
1"]))

    return fpr, tpr, roc_auc, cm

# Εκτέλεση αξιολόγησης (Τα 6 Τελικά Μοντέλα μας)
fpr_rf, tpr_rf, auc_rf, cm_rf = evaluate_model_full(df_rf, "Random Forest",
"probability")
fpr_svm, tpr_svm, auc_svm, cm_svm = evaluate_model_full(df_svm, "SVM",
"rawPrediction")
fpr_nb, tpr_nb, auc_nb, cm_nb = evaluate_model_full(df_nb, "Naive Bayes",
"probability")
fpr_mlp, tpr_mlp, auc_mlp, cm_mlp = evaluate_model_full(df_mlp, "Neural
Networks", "probability")
fpr_gbt, tpr_gbt, auc_gbt, cm_gbt = evaluate_model_full(df_gbt, "GBT Classifier",
"probability")
fpr_lr, tpr_lr, auc_lr, cm_lr = evaluate_model_full(df_lr, "Logistic Reg.",

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"probability")

# =====
# 1. ΠΙΝΑΚΑΣ ΣΥΓΚΡΙΣΗΣ (Κεφάλαιο 4: Model Evaluation)
# =====
print("\n" + "="*80)
print("ΠΙΝΑΚΑΣ ΣΥΓΚΡΙΣΗΣ ΜΟΝΤΕΛΩΝ (Απαίτηση Εκφώνησης)")
print("="*80)
comparison_df = pd.DataFrame(metrics_dict).round(3)
print(comparison_df.to_string(index=False))
print("="*80)

# =====
# 2. ΟΠΤΙΚΟΠΟΙΗΣΗ 1: CONFUSION MATRICES (Τέλειο 2x3 Grid)
# =====
fig, axes = plt.subplots(2, 3, figsize=(16, 10))
fig.suptitle('Confusion Matrices (Απαίτηση Εκφώνησης)', fontsize=18,
fontweight='bold')

cms = [cm_rf, cm_svm, cm_nb, cm_mlp, cm_gbt, cm_lr]
titles = ['Random Forest', 'SVM', 'Naive Bayes', 'Neural Networks', 'GBT
Classifier', 'Logistic Reg.']
colors = ['Blues', 'Oranges', 'Greens', 'Purples', 'Reds', 'YlGnBu']

for i in range(len(colors)):
    row, col = i // 3, i % 3
    sns.heatmap(cms[i], annot=True, fmt='d', cmap=colors[i], cbar=False,
ax=axes[row, col])
    axes[row, col].set_title(titles[i], fontsize=12)
    axes[row, col].set_xlabel('Predicted')
    axes[row, col].set_ylabel('True')

plt.tight_layout()
plt.subplots_adjust(top=0.9)
plt.show()

# =====
# 3. ΟΠΤΙΚΟΠΟΙΗΣΗ 2: ROC CURVES
# =====
plt.figure(figsize=(11, 8))
plt.plot(fpr_rf, tpr_rf, label=f'Random Forest (AUC = {auc_rf:.3f})', lw=2)
plt.plot(fpr_svm, tpr_svm, label=f'SVM (AUC = {auc_svm:.3f})', lw=2)
plt.plot(fpr_nb, tpr_nb, label=f'Naive Bayes (AUC = {auc_nb:.3f})', lw=2)
plt.plot(fpr_mlp, tpr_mlp, label=f'Neural Networks (AUC = {auc_mlp:.3f})', lw=2)
plt.plot(fpr_gbt, tpr_gbt, label=f'GBT (AUC = {auc_gbt:.3f})', lw=2,
linestyle='--')
plt.plot(fpr_lr, tpr_lr, label=f'Logistic Reg. (AUC = {auc_lr:.3f})', lw=2,

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linestyle='--')
plt.plot([0, 1], [0, 1], 'k--', lw=2, label='Random Chance')
plt.title('Σύγκριση Μοντέλων: Καμπύλη ROC', fontsize=16, fontweight='bold')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate (Recall)')
plt.legend(loc="lower right")
plt.grid(alpha=0.3)
plt.show()

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# 4. ΟΠΤΙΚΟΠΟΙΗΣΗ 3: ΣΥΓΚΡΙΤΙΚΑ BAR CHARTS
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# =====
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bar_df = comparison_df.drop(columns=["ROC-AUC"])
metrics_melted = bar_df.melt(id_vars="Model", var_name="Metric",
value_name="Score")

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plt.figure(figsize=(14, 7))
sns.barplot(x='Model', y='Score', hue='Metric', data=metrics_melted,
palette='Dark2')
plt.title('Συγκριτικό Διάγραμμα Μετρικών (Accuracy, Precision, Recall, F1-
Score)', fontsize=16, fontweight='bold')
plt.ylim(0, 1.1)
plt.ylabel('Score')
plt.xticks(rotation=30)
plt.legend(title='Μετρικές', bbox_to_anchor=(1.01, 1), loc='upper left')
plt.grid(axis='y', alpha=0.3)
plt.tight_layout()
plt.show()

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```
spark.stop()
```

Εκκίνηση SparkSession...

Φόρτωση προβλέψεων (Pandas)...

```
===== Random Forest =====
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	precision	recall	f1-score	support
Class 0	0.98	0.79	0.88	945
Class 1	0.12	0.67	0.21	42
accuracy			0.78	987
macro avg	0.55	0.73	0.54	987
weighted avg	0.95	0.78	0.85	987

```
===== SVM =====
```

	precision	recall	f1-score	support
Class 0	0.98	0.74	0.84	945
Class 1	0.10	0.64	0.17	42

accuracy			0.74	987
macro avg	0.54	0.69	0.51	987
weighted avg	0.94	0.74	0.82	987

===== Naive Bayes =====

	precision	recall	f1-score	support
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Class 0	0.99	0.40	0.57	945
Class 1	0.06	0.88	0.11	42

accuracy			0.42	987
macro avg	0.52	0.64	0.34	987
weighted avg	0.95	0.42	0.55	987

===== Neural Networks =====

	precision	recall	f1-score	support
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Class 0	0.98	0.75	0.85	945
Class 1	0.10	0.62	0.17	42

accuracy			0.74	987
macro avg	0.54	0.68	0.51	987
weighted avg	0.94	0.74	0.82	987

===== GBT Classifier =====

	precision	recall	f1-score	support
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Class 0	0.98	0.77	0.86	945
Class 1	0.12	0.71	0.21	42

accuracy			0.77	987
macro avg	0.55	0.74	0.54	987
weighted avg	0.95	0.77	0.84	987

===== Logistic Reg. =====

	precision	recall	f1-score	support
--	-----------	--------	----------	---------

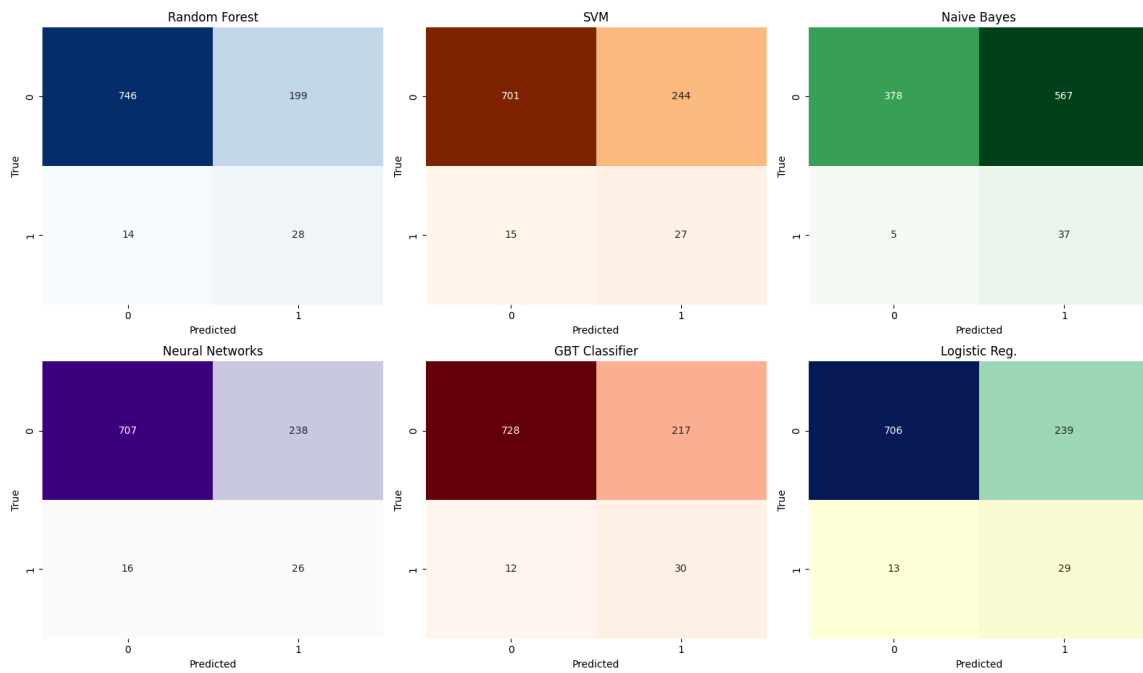
Class 0	0.98	0.75	0.85	945
Class 1	0.11	0.69	0.19	42

accuracy			0.74	987
macro avg	0.55	0.72	0.52	987
weighted avg	0.94	0.74	0.82	987

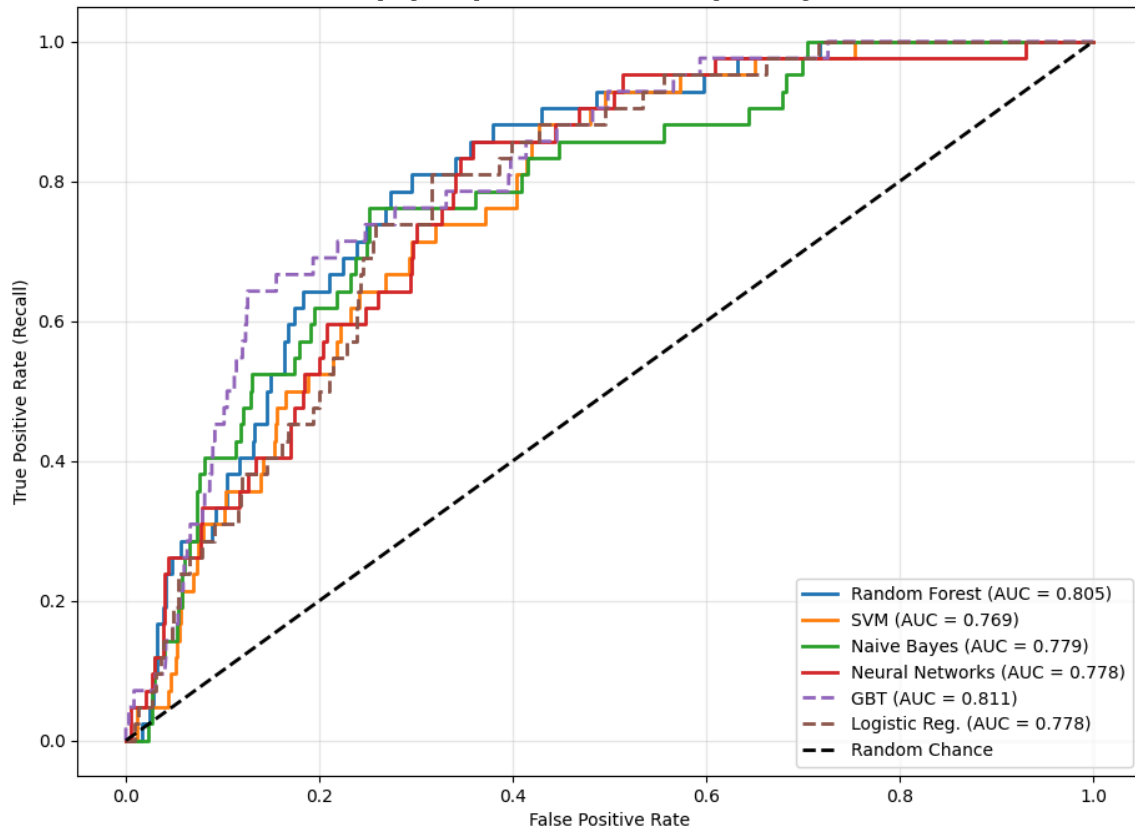
ΠΙΝΑΚΑΣ ΣΥΓΚΡΙΣΗΣ ΜΟΝΤΕΛΩΝ (Απαίτηση Εκφώνησης)

Model	Acc (Total)	P (Class 0)	R (Class 0)	F1 (Class 0)	P (Class 1)
R (Class 1)	F1 (Class 1)	ROC-AUC			
Random Forest	0.784	0.982	0.789	0.875	0.123
0.667	0.208	0.805			
SVM	0.738	0.979	0.742	0.844	0.100
0.643	0.173	0.769			
Naive Bayes	0.420	0.987	0.400	0.569	0.061
0.881	0.115	0.779			
Neural Networks	0.743	0.978	0.748	0.848	0.098
0.619	0.170	0.778			
GBT Classifier	0.768	0.984	0.770	0.864	0.121
0.714	0.208	0.811			
Logistic Reg.	0.745	0.982	0.747	0.849	0.108
0.690	0.187	0.778			

Confusion Matrices (Απαίτηση Εκφώνησης)



Σύγκριση Μοντέλων: Καμπύλη ROC



Συγκριτικό Διάγραμμα Μετρικών (Accuracy, Precision, Recall, F1-Score)

